IMA FAST TRACK
6 MONTHS A/L
ENGLISH MEDIUM

Economics

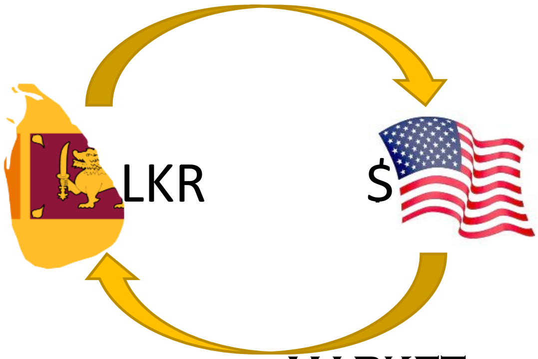
International Finance

KRISHAN RATHNAYAKE

GCE ADVANCED LEVEL ENGLISH MEDIUM

Foreign exchange

Foreign exchange, or forex, is the conversion of one country's currency into another.



Foreign exchange MARKET

Relations of institutional structures where foreign exchange is exchanged is defined as foreign exchange market.

In other words intuitional structures where the amount of foreign exchange is determined by demand for foreign currency and supply of foreign currency is defined as foreign exchange market.

- Foreign exchange market has organized to facilitate international trade and financial transactions by providing foreign exchange.
- Foreign exchange markets are considered as largest markets today

Various participants of a foreign exchange market.

- Commercial banks
- Foreign exchange brokers
- Business firms
- Central bank
- General public
- Government

Commercial banks perform a major role within foreign exchange market. In this publishing of purchasing price and selling price of foreign exchange is important. The gap between selling price and purchasing price shows the profit margin of a commercial bank

Profiteers Behavior

If there is any change of prices occur at the centers of main foreign exchange market profiteers of foreign exchange takes advantage and earn profit s of it.

WHICH MEANS THEY PURCHASE FOREIGN EXCHANGE FORM THE MARKET WHERE THE PRICE IS LOW AND SELL THEM AT THE FOREIGN EXCHANGE MARKET WHERE THE PRICE IS HIGH.

Nominal exchange rate

The rate at which a unit of domestic currencies converted into unit of foreign currency or the rate at which unit of foreign currency is converted into a unit of domestic currency is known as foreign exchange rate.

In this way expressing the price of one currency in terms of another currency is known as nominal exchange rate.

Two ways of expressing foreign exchange rate.

Direct quotation

Expressing the value of foreign currency in terms of domestic currency is the direct quotation system or the price quotation system.

EX- showing of how many rupees are needed to purchase one US dollar.

US $ 1 = SLKR 152

Indirect quotation

Expressing the value of domestic currency in terms of foreign currency is the indirect quotation system or the volume quotation system.

The reciprocal of the direct quotation, of foreign exchange is the indirect quotation of foreign exchange rate.

Ex- the rupee value of a us dollar is determined by the demand and supply of us dollar at the foreign exchange market.

152 1 1SLKR = US$ = 0.0065152

Demand & Supply of Exchange Rates

Us dollars are demanded to settle payment transactions of balance of payments such as

- to import goods
- travel to other countries,
- to repay debt,
- to pay interest for borrowings
- to pay transfers to foreign countries.

There is an inverse relationship exists between rupee value of us dollar and the quantity of dollars demanded.

Dollar demand curve will shift to the right on the situations of where

- imports increase at the prevailing exchange rate
- foreign tours increase
- interest payment for debt increase
- transfer payments to foreign countries increase.

Dollar demand curve shift to the left at the situations where

- imports decrease,
- foreign tours decrease
- transfer payments decrease.

Supply of dollars determined by the receipts transactions of the balance payments.

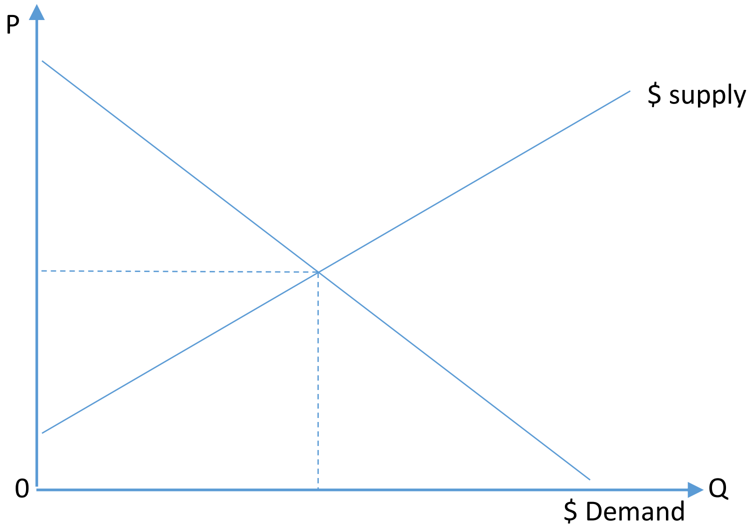
- such as exports of goods,
- receipt of primary income
- receipts of secondary income.

A positive relationship exists between rupee value of dollar and supply of dollar.

When exports receipts of tourism and loans increase at the prevailing exchange rate system supply of dollars increase doller supply curve shifts to the right .

When exports,, receipts of services decrese, supply of dolloars also decrese dollar supply curve shifts to the left.

Exchange Rate Market equlibrium



Nominal Exchange Rates

____

____

____

____

____

____

Exchange rate systems.

- Fixed exchange rate system
- Floating exchange rate system

____

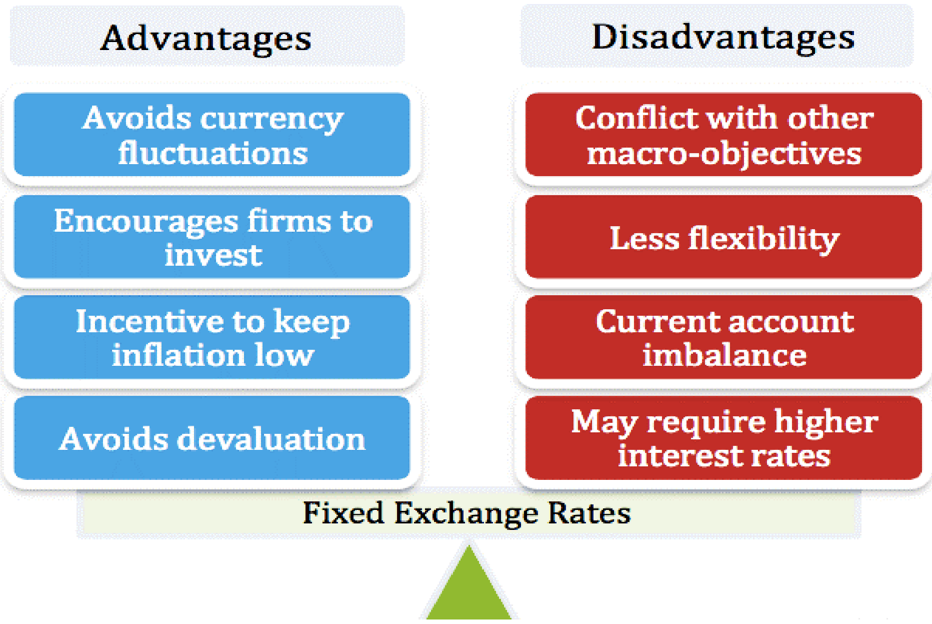
____

____

◆ Fixed exchange rate system

If exchange rate determined by the monetary authority prevails at the same level till it exchange by the same monetary authority it is known as fixed exchange rate system.

- Fixed exchange rate system was implemented at many countries of the world during the period of 1890 to 1920



Gold standard was the basis for fixed exchange range system. Which means issuing of currency based on a gold reserve system. When USA deposited one ounce of gold at Federal Reserve Bank it issued 35 US dollars. When one ounce of gold deposited at Central Bank of England it issued 12.50 Sterling pounds. Following this the exchange rate between US \$ and £ was 1 £ = US \$ 2.80. This value remained constant until the amount of dollars or sterling pounds issued against one ounce of gold changed.

◆ <u>Floating exchange rate system</u>

Floating exchange rate system would generate a mechanism which will automatically balance of payments.

When there is a deficit in balance of payments exchange rate will start to increase as demand for foreign exchange increase and supply of foreign exchange decrease. Due to that external value of money decrease and as a result exports are encouraged and imports are discouraged. It will crates balance of payments equilibrium. On the other hand when there is a surplus in balance of payments supply of foreign exchange increases and demand decreases. Due to that exchange rate will start to decrease and as a result external value of money will start to increase. In this situation imports increase and exports and exports decrease and due to that balance of payments will reach its equilibrium.

Advantages of a floating exchange rate system

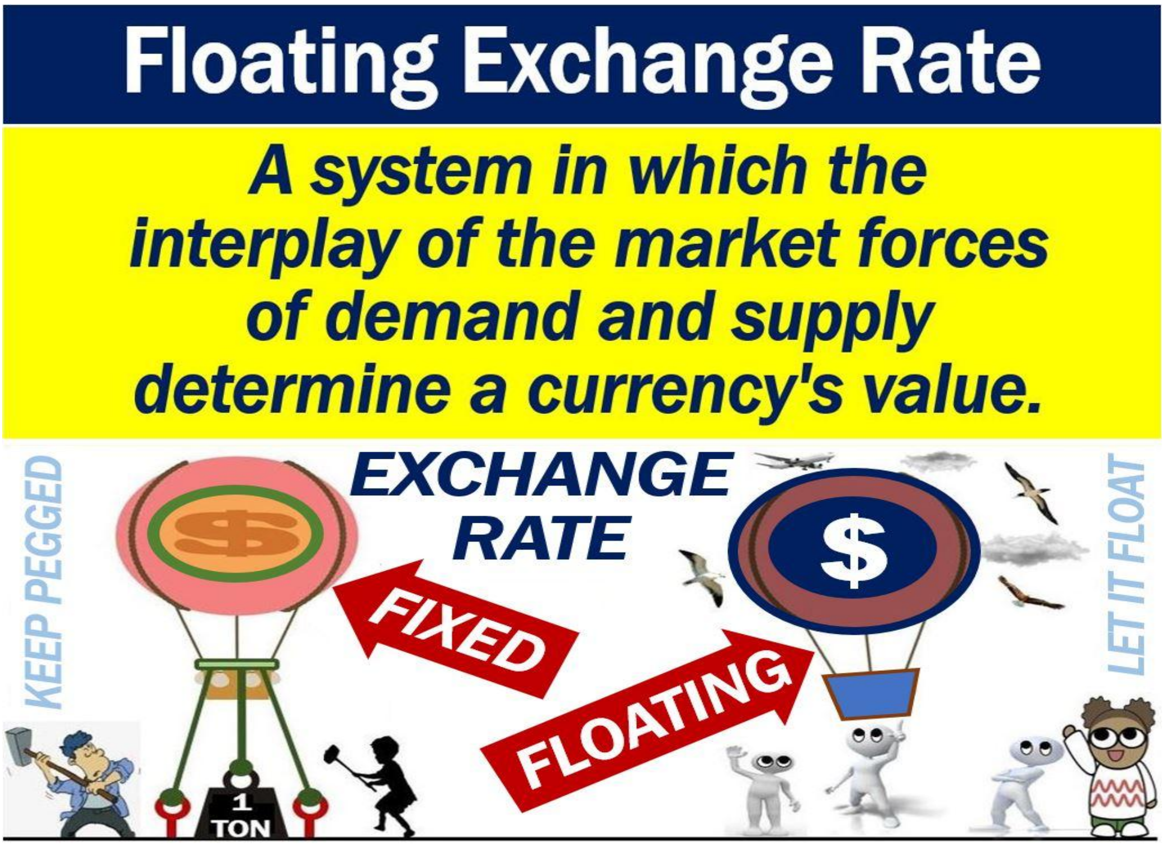
- As floating exchange rate system is a system which equate demand for foreign exchange and supply of foreign exchange there is no room for exchange rate over valuation or devaluation.
- Global economic resources would distribute among alternative uses more efficiently as the exchange rate being more realistic.
- As floating exchange rate system creates external stability without any influence policy makes can concentrate on internal stability
- When determination exchange rate as demand and supply at the foreign exchange market being equal there should not be any deficit or surplus situation arise. Because of this there is no necessity to maintain foreign services.
- Also as protectionist policies should not be followed to control imports and exports more opportunities will have to maximum social welfare through free trade

Disadvantages of a floating exchange rate system are given below.

- Floating exchange rate system will be barrier to promote international trade as it is always subjected to fluctuations.
- Floating exchange rate system would generates uncertainties over foreign investors and lenders.
- As speculation to foreign exchange market activities are common with floating exchange rate system it will create more fluctuations within exchange rate.

Floating versus Fixed Exchange Rates

Floating Currency	Fixed Currency
• Reduces the need for foreign currency reserves • Freedom to set policy interest rates to meet domestic objectives • May help to prevent imported inflation • Insulation for an economy after an external shock especially for export-dependent countries • Partial automatic correction for a current account deficit • Less risk of speculative attack	• Certainty of currency value gives confidence for inward investment • Reduced costs of currency hedging for businesses • Stability helps to control inflation – it is a discipline on businesses to keep unit labour costs low and drive efficiency • Can lead to lower borrowing costs (lower yields on bonds) • Imposes responsibility on government policies • Less speculation if the fixed exchange rate is credible



◆ Managed floating exchange

Managed floating exchange rate system is a combination of both fixed exchange rate and floating exchange rate. It is constructed to given benefits of both fixed exchange rate and floating exchange rate.

• Maintaining of a fixed exchange rate would provide a room to more exchange rate between a particular range.

If exchange rate change beyond this range monetary authority intervention to the foreign exchange market and if it control fluctuations by buying and selling domestic currency at the foreign exchange market then the system is known as managed floating exchange rate system.

• In this way, although monetary authority intervenes to the foreign market in short term it is expected to reach a long term stability.

• Exchange rate prevails for the currency of a particular country change in various ways and in various quantities relative to various foreign currencies, to calculate mean value of those changes effective exchange rate is calculated.

• There are two forms of effective exchange rates.

◆ Nominal effective exchange rate

Nominal effective exchange rate is the unadjusted average rate which determines on the basis of Sri Lanka's 24 major trading partners and which estimated on the weight given to each country based on the size of foreign trade takes place between countries.

◆ Real effective exchange rate

Real effective exchange rate is the exchange rate made adjusting average change of Sri Lanka's inflation rate and other trading partners inflation rate to the nominal effective exchange rate

Relationship between nominal exchange rate and real exchange rate

Trading Partners	Nominal exchange rate (R)	Weight given on international trade (w)	$R \times W$	Sri Lanka's inflation rate	Foreign countries inflation rate	Change
1. USA	1 \$ = Rs. 150	30	4500	8	10	-2
2. UK	1 \$ = Rs. 200	20	4000	8	12	-4
3. India	1 Rs = Rs.225	50	112.50	8	-5	+3
4. China	1 U = Rs.2	10	20	8	10	-2
5. Japan	1 F = Rs.2	5	10	8	5	+3
		$\Sigma W = 100$	8580 ΣRW			-8+6=-2

$$\text{Nominal Effective Exchange Rate} = \frac{E_{RW}}{E_W}$$

$$= \frac{8580}{100}$$

$$= \underline{\underline{85.80}}$$

Nominal effective exchange rate 85.80 means in any country mentioned above which engage in foreign trade with Sri Lanka average exchange rate is equal to 85.80. After adjusting average change of between countries inflation to nominal effective exchange rate the value of real effective exchange is obta l.

Real effective exchange rate = Nominal effective exchange rate + Average change of inflation between countries

= 85.80 + (-2)

= 84.42

changes of Foreign Exchange Rate and its consequences

Exchange rate can be changed on demand for foreign currency and supply of foreign currency.

This happens under a floating exchange rate system.

Example : Commercial banks daily display the values of various exchange rates.

Based on changes of such a display purchasing and selling of foreign exchange of an economy takes place.

- Devaluation and revaluation of exchange rate happens within fixed exchange rate system.
- Depreciation and appreciation of exchange rate happens within floating exchange rate system.

Several factors reasons affect for change in foreign exchange rate

- International trade
- Capital mobility
- Price fluctuations
- Speculation
- Economic strength
- Government policies
- Share market activities
- Political Situation

Devaluation and depreciation

Lowing of exchange rate than official exchange rate at a situation of continuous deficit in balance of payments is known as devaluation.

Decreasing the value of domestic currency in terms of a foreign currency is called depreciation. Depreciation of foreign exchange rate system happens under a floating exchange rate system.

The economic effects of devaluation of exchange rate

- Increase in the domestic price of imports
- Increase in import expenditure than before
- Domestic inflation can be increased if domestic demand depends upon imports.
- increase in foreign debt service payment.
- Balance of service account of balance of payments will be favorable due to increase in the earnings from tourism.
- Secondary income account being favoruable due to increase in worker remittance including middle east employees remittances.
- Import demand directed towards domestic goods due to discouraging of imports with an increase in price of imports.
- Export earnings can be increased due to increase in demand for exports.

Example : Assume value of one us dollar as Rs. 100 and the value of one unit if Sri Lankan textile export as Rs 100. If one us dollar value increased to Rs. 150 due to exchange rate depreciation then price of one unit of textileexport will be 0.66 dollars. Due to this demand for exports increases

Revaluation and appreciation

Increase in the price of domestic currency in terms of foreign currency is known as revaluation of foreign exchange rate.

Example : if value of one us dollar is Rs. 114 following direct quotation system and if this value decrease to Rs. 100 it shown an increase in the value of exchange rate. At this situation due to increase in the value of rupee in terms of dollar more dollars should be paid to purchase rupees.

Under a floating exchange rate system the value of exchange rate can be appreciated. However, the effects of both revaluation and appreciation are similar.

The economic effects of exchange rate appreciation

- Decrease in the domestic prices of imports.
- Decrease in inflation rate.(If defend upon imports)

The value of foreign debt interms of domestic currency decreases.

- Decrease in the cost of repaying foreign debt.

Objectives of foreign exchange devaluation.

- minimizing of long term deficit in balance of payments.
- Discouraging of import demand
- Encouraging of export supply
- Encouraging of worker remittances
- Improving of foreign direct investments

Some conditions should be fulfiller in order to achieve above objectives

I. Foreign demand price for exports should be elastic

II. Domestic demand price for imports should be elastic

III. Sum value of price elasticity of demand for exports and imports being greater than one (Marshall-Lerner condition)

IV. Domestic supply related to exports being elastic

V. Domestic price level remains constant

Balance of Payments

The balance of payments (BOP), also known as the balance of international payments, is a statement of all transactions made between entities in one country and the rest of the world over a defined period, such as a quarter or a year.

It is a schedule that states the form of balancing the changes between foreign currency which gain from foreign transactions and the foreign currency which pay for foreign countries when a country is is engage in economic transactions with the rest of the world during a period of year.

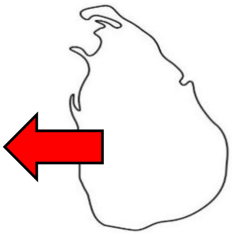
In this all economic transactions which occur between residents and non-residents is recorded systematically according to the principle of double entry system.

• The all transactions of balance of payments sheet should be reported (recorded) in two places with the same value according to the basic standard.

• The value of all transactions must b e note down in two columns as credit and debit and the value of debit must be reported as (+) and the value of credit must be reported as(-).

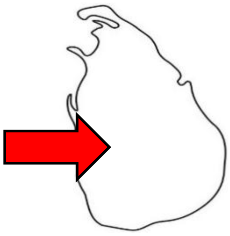
• The net balance of all transactions would be zero when the value of all transactions are noted in a credit column and a debit column with a same value. This is the meaning of "Balance of payments always being the balanced.

• The two principles should be followed when recording the values of credit and debit separately



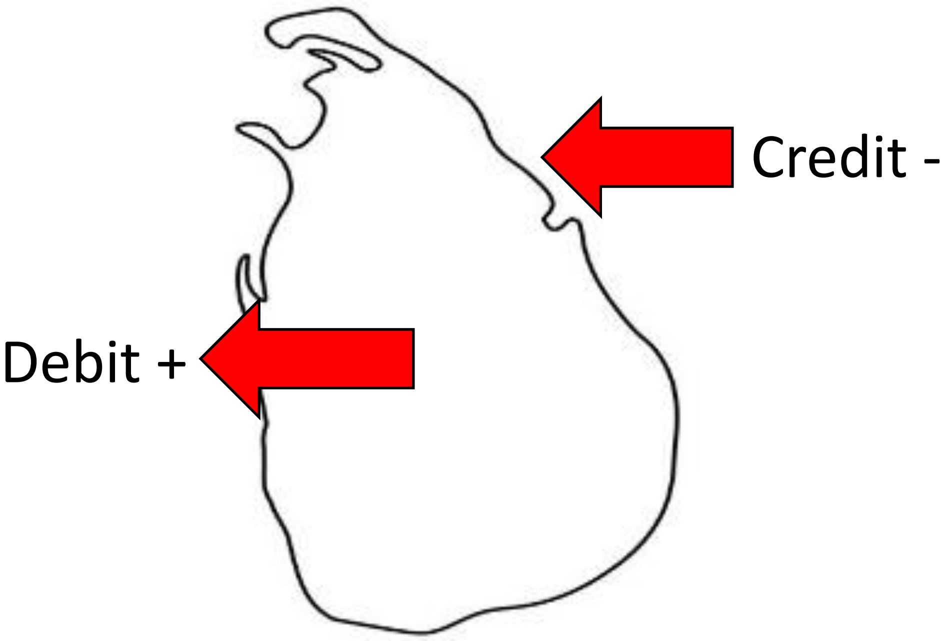
It should be reported the value of item as a debit value (+) when flow of an asset / a good / or a service flows from the country to another country.

Example : When Sri Lanka export garments it should be included to current account as a debit value.



It should be reported the value of item as a credit value (-) when receive a good / a service or an asset receive to the country from a another country.

Example : When Sri Lanka imports fuel from a middle east country that value should be included in current account as a credit value.



✪ Balance of payment transactions are 5 types according to the new structure.

1. The transactions related to goods and services exports and imports.

2. The transactions related to income receipts and payments

3. The one way transactions related to the things such as assets, wealth and gifts.

4. The transactions related to buying and selling of financial assets .

5. The transactions related to buying and selling of non-produced and non financial assets

New structure of balance of payments

• Current account

•Capital account

•Financial account

1 Goods and services account

1.1 Goods Account (Trade)

1.2 Services Account

2. Primary Income Account

3. Secondary Income Account

Current Account

❑ Goods account (Trade account)

The account which is reported the exporting and importing of goods with rest of the world is known as Trade Account.

• The non monetary gold of trade account is reported as a store of value of private sector

The balance of trade account is known as balance of trade. It is also known as the balance of visible goods account.

Balance of trade = Goods exports – imports

v

Exports 500,

Imports = 1000 (million)

Balance of trade = Exports of goods – Imports goods

= 500 – 1000

= - 500

1- Exports 500, Imports = 1000 (million) Calculate country's trade account balance

2- Exports 100, Imports = 20 (million) Calculate country's trade account balance

3- Exports 500, Imports = 200 (million) Calculate country's trade account balance

4- Exports 2000, Imports = 1200 (million) Calculate country's trade account balance

5- Exports 400, Imports = 700 (million) Calculate country's trade account balance

O Services Account

The account which is reported expenditures and incomes of non factor services of a country with rest of the world in the international trade is known as services account.

• Following components are included for services account.

- Travel

- Sea Transport
- Air transport
- Financial Services
- Telecommunications services
- Insurance Services

Balance of services accounts = Export income of services – Import expenditure of services

1- Exports services 500, Imports services = 200 (million) Calculate country's services account balance

2- Exports services 50, Imports services = 20 (million) Calculate country's services account balance

3- Exports services 750, Imports services =1000 (million) Calculate country's services account balance

4- Exports services 300,Imports services =250 (million) Calculate country's services account balance

5- Exports services 800, Imports services = 700 (million) Calculate country's services account balance

Goods & Services Account

The balance of goods and services account can be calculated by adjusting the balance of goods account to balance of services account.

Balance of goods and services account = Balance of goods account + Balance of services account.

• Some examples related to foreign transactions of Sri Lanka are given below.

Balance of goods account = 2000

Balance of services account= 3000

Calculate the goods and services account

Balance of goods and services account = Balance of goods account + Balance of services account.

=

2) Primary Income Account

Balance of primary income account = Foreign factor income received – Foreign factor income paid

The items included to primary income account are interest, wages, dividends obtain by investing in financial assets which arise from the production process.

• Balance of primary income account = foreign factor income received – Foreign factor income paid

= 300 – 500

= - 200

3) Secondary Income Account

All kind of foreign transfers Receipts foreign transfers payments can be recorded under the secondary income.

Sri Lanka who works at middle east sent Rs. 1500 million of workers' remittances to commercial banks.

Chinese who live in Sri Lanka sent Rs. 500 million of workers at middle east sent Rs. 1500 million of workers’ remittances to a bank in china.

Secondary income account = Receipts of foreign transfers– payments of foreign transfers.

=1000 - 1500

= - 500

- It can be gain the balance of current account by matching the balances of good and services account, primary income account and secondary income.

	Debit	**Credit**	**Balance**
Balance of goods and services account	500	1000	-500
Balance of primary income account	300	500	-200
Balance of secondary income account	1500	500	500
Balance of current account			-200

Ex 1 -

Goods exports = 200

Goods imports= 180

Services Exports= 400

Services imports=300

Foreign factor income received= 500

Foreign factor income paid=200

Receipts of foreign transfers=1000

payments of foreign transfers=1980

1- Calculate trade account balance

____

____

2- Calculate the balance of services account

____

____

3- Calculate the balance of goods and services account

____

____

____

4- Calculate primary income balance

____

____

____

5- Calculate secondary income balance

____

____

____

6- Caulate current account balance

____

____

____

Ex 2 -

Goods exports = 1200

Goods imports= 1180

Services Exports= 1400

Services imports=1300

Foreign factor income received= 1500

Foreign factor income paid=1200

Receipts of foreign transfers=11000

payments of foreign transfers=11980

1- Calculate trade account balance

____

____

____

2- Calculate the balance of services account

____

____

____

3- Calculate the balance of goods and services account

____

____

____

4- Calculate primary income balance

5- Calculate secondary income balance

6- Caulate current account balance

Ex 3 -

Goods exports = 400

Goods imports= 600

Services Exports= 500

Services imports=200

Foreign factor income received= 2000

Foreign factor income paid=1000

Receipts of foreign transfers= 3000

payments of foreign transfers=2000

1- Calculate trade account balance

2- Calculate the balance of services account

3- Calculate the balance of goods and services account

4- Calculate primary income balance

5- Calculate secondary income balance

6- Caulate current account balance

Ex 4 -

Goods exports = 1200

Goods imports= 1180

Services Exports= 1400

Services imports=1300

Foreign factor income received= 1500

Foreign factor income paid=1200

Receipts of foreign transfers=11000

payments of foreign transfers=11980

1- Calculate trade account balance

2- Calculate the balance of services account

3- Calculate the balance of goods and services account

4- Calculate primary income balance

____
____
____

5- Calculate secondary income balance

____
____
____

6- Caulate current account balance

____
____
____

Ex 5 -

Goods exports = 1250

Goods imports= 1080

Services Exports= 1300

Services imports=1250

Foreign factor income received= 1400

Foreign factor income paid=1200

Receipts of foreign transfers=5500

payments of foreign transfers=4500

1- Calculate trade account balance

____
____
____

2- Calculate the balance of services account

____
____
____

3- Calculate the balance of goods and services account

____
____
____

4- Calculate primary income balance

____

____

____

5- Calculate secondary income balance

____

____

____

6- Caulate current account balance

____

____

____

Ex 6 -

Goods exports = 300

Goods imports= 600

Services Exports= 450

Services imports=250

Foreign factor income received= 1000

Foreign factor income paid=800

Receipts of foreign transfers= 6000

payments of foreign transfers= 7000

1- Calculate trade account balance

____

____

____

2- Calculate the balance of services account

____

____

____

3- Calculate the balance of goods and services account

____

____

____

4- Calculate primary income balance

5- Calculate secondary income balance

6- Caulate current account balance

Ex 7 -

Goods exports = 1300

Goods imports= 2600

Services Exports= 1450

Services imports=850

Foreign factor income received= 2000

Foreign factor income paid=1800

Receipts of foreign transfers= 5000

payments of foreign transfers=4000

1- Calculate trade account balance

2- Calculate the balance of services account

3- Calculate the balance of goods and services account

4- Calculate primary income balance

5- Calculate secondary income balance

6- Caulate current account balance

Ex 8 -

Goods exports = 500

Goods imports= 600

Services Exports= 3000

Services imports= 2900

Foreign factor income received= 3000

Foreign factor income paid=2000

Receipts of foreign transfers= 4000

payments of foreign transfers=2000

1- Calculate trade account balance

2- Calculate the balance of services account

3- Calculate the balance of goods and services account

4- Calculate primary income balance

____
____
____

5- Calculate secondary income balance

____
____
____

6- Caulate current account balance

____
____
____

Capital Account

Non – produced non-monetary assets are included in capital

The non-produced and non monetary assets between residents and non-residents are included in capital account as credit and debit.

Examples for non-produced monetary assets

- Licence
- Lease agreements
- Brand names
- Buying and selling of goodwill
- Selling a Land to an embassy
- Donations from governments and institutions of foreign countries
- Patent
- Royalties

Balance of capital account= Receipts - Payments

Ex 01-

Capital receipts = 400

Capital payments=200

Calculate the balance of capital account

____
____

____
____

Ex 02-

Capital receipts = 500

Capital payments=300

Calculate the balance of capital account

____
____
____
____

Ex 03-

Capital receipts = 800

Capital payments=500

Calculate the balance of capital account

____
____
____
____

Ex 04-

Capital receipts = 1000

Capital payments=800

Calculate the balance of capital account

____
____
____
____

Ex 05-

Capital receipts = 1500

Capital payments=500

Calculate the balance of capital account

____
____
____
____

Financial Account

The financial account is reported the transactions related to monetary assets and liabilities that take place between residents and non-residents.

The transactions of financial account can be divided mainly five types based on its function.

- Direct investments
- Portfolio investments
- Financial derivatives
- Other investments
- Foreign Reserve Assets